

Compact space.

1. a). Let  $\mathcal{C}, \mathcal{C}'$  be Topologies on  $X$ , with  $\mathcal{C} \subseteq \mathcal{C}'$ .

If  $(X, \mathcal{C}')$  is compact, then  $(X, \mathcal{C})$  is compact.

Proof. Let  $\mathcal{O} \subseteq \mathcal{C}$  be an open cover of  $X$ . Then, since  $\mathcal{O} \subseteq \mathcal{C}'$  and  $(X, \mathcal{C}')$  compact,  $\mathcal{O}$  has finite subcover,  $\square$

1. b). If  $X$  is Compact  $T_2$  under both  $\mathcal{C}$  and  $\mathcal{C}'$ ,

then either  $\mathcal{C}$  and  $\mathcal{C}'$  are equal or they are Not comparable.

Proof. If  $\mathcal{C} = \mathcal{C}'$ , then there is nothing to prove. Suppose  $\mathcal{C} \neq \mathcal{C}'$ .

Enough to show that neither  $\mathcal{C} \subseteq \mathcal{C}'$  nor  $\mathcal{C}' \subseteq \mathcal{C}$ .

Suppose that  $\mathcal{C} \subseteq \mathcal{C}'$ . Then, we can choose an open  $U \in \mathcal{C}' \setminus \mathcal{C}$ .

Now,  $X \setminus U$  is closed in  $\mathcal{C}'$  but not  $\mathcal{C}$ . Since  $\mathcal{C}'$  is Compact  $T_2$ ,

$X \setminus U$  is Compact set of  $\mathcal{C}'$ . Since  $\mathcal{C} \subseteq \mathcal{C}'$ , for any open cover  $\mathcal{O} \subseteq \mathcal{C}$  for  $X \setminus U$ ,

it has finite subcover. Therefore,  $X \setminus U$  is compact in  $(X, \mathcal{C})$ , thus closed.

But, it is contradiction. Similarly,  $\mathcal{C}' \not\subseteq \mathcal{C}$ .

2. a). Every subspace in Co-finite Topology  $(\mathbb{R}, \mathcal{C}_f)$  is compact.

b). In Co-Countable  $(\mathbb{R}, \mathcal{C}_c)$ , is  $[0, 1]$  compact? an

Proof. Let  $A \subseteq \mathbb{R}$  be a subset. Let  $\mathcal{O} = \{\mathbb{R} \setminus F \mid F \text{ is finite}\}$  be an open cover for  $A$ . There is,

$$A \subseteq \bigcup_{U \in \mathcal{O}} U \Leftrightarrow \bigcap_{U \in \mathcal{O}} \mathbb{R} \setminus U \subseteq A^c$$

But since  $\mathbb{R} \setminus U$  is finite for each  $U \in \mathcal{O}$ , there are finite number of  $U_i, \dots, U_n$  s.t.

$$\bigcap_{i=1}^n \mathbb{R} \setminus U_i \subseteq A^c \Leftrightarrow A \subseteq \bigcup_{i=1}^n U_i$$

Therefore  $A$  is Compact. (More specifically, for any distinct finite set  $F_1$  and  $F_2$ ,  $F_1 \cap F_2$  has number of less than  $\min\{\#(F_1), \#(F_2)\}$ .)

But, in the Co-Countable,  $[0, 1]$  is Not compact, because

